# MAGNETIC GAP MEASUREMENT AND DISPLACEMENT SENSING EXPANSION

## Magnetic Displacement Measurement Concept

In certain embodiments, the wearable longitudinal penile physiology monitoring device may incorporate magnetic field–based displacement sensing to measure relative motion between structural regions of the wearable. Magnetic sensing may enable detection of expansion, compression, stiffness response, and deformation behavior associated with physiologic state transitions without requiring direct measurement of internal tissue pressure.

Magnetic displacement measurement may be implemented using one or more magnetic elements and one or more magnetic field sensors configured to detect changes in field strength, vector orientation, or spatial magnetic distribution resulting from structural movement.

## Hall Effect Sensing Embodiments

A Hall effect sensor may be positioned on a first structural arm of an open-gap wearable while a permanent magnet is positioned on an opposing arm. Relative displacement between the arms alters magnetic field intensity measured by the sensor, enabling inference of gap distance, mechanical deformation, or rigidity-related expansion. Multiple Hall sensors or multi-axis magnetic sensors may be employed to improve linearity, dynamic range, or orientation independence.

Magnet geometries may include cylindrical magnets, rectangular magnets, ring magnets, segmented magnet arrays, or distributed magnetic materials integrated into structural components. Magnetic polarity orientation, shielding structures, and calibration methods may be varied without departing from the scope of the invention.

## Alternative Magnetic and Non-Magnetic Displacement Methods

Magnetic sensing may alternatively employ magnetoresistive sensors, fluxgate sensors, inductive sensing, reed switches, or other magnetic detection technologies. In further embodiments, displacement measurement may be achieved through capacitive distance sensing, optical distance sensing, ultrasonic ranging, resistive displacement measurement, strain-based gap inference, or mechanical linkage monitoring systems capable of detecting relative motion.

Displacement sensing methods may operate independently or in combination with additional rigidity sensing modalities to improve robustness against noise, misalignment, or individual anatomical variability.

## Calibration and Compensation Techniques

Calibration procedures may compensate for temperature variation, magnetic drift, mechanical tolerance differences, or user-specific fit characteristics. Calibration may occur during initial device setup, periodically during operation, or continuously through adaptive computational learning algorithms.

## Distributed Magnetic Architectures

Multiple magnetic sensing regions may be distributed around the wearable device to capture multi-point deformation patterns or asymmetric expansion behavior. Distributed magnetic architectures may enable enhanced rigidity modeling, improved signal redundancy, and spatial mapping of mechanical response.

The magnetic sensing embodiments described herein are illustrative and non-limiting and encompass present and future displacement sensing technologies capable of measuring wearable deformation associated with longitudinal penile physiology monitoring.